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PRACTITIONER'S REPORT

A review of proficiency testing scheme in Republic of Korea: in the field of water analysis

Young-Hee Kim · Jung-Sub Lee · Ki-Bong Song · Young-Yeul Kang · Sun-Kyoung Shin · Gi-Taeg Jung · Eun-Jin Hong · Hyun-Jeong Kim · Kum-Hee Kim

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Abstract For regulatory purposes, it is often required to check the quality of environmental chemical analyses by routine proficiency testing (PT). In this study, the PT scheme of environmental laboratories in Republic of Korea has been reviewed. Especially, real water-based reference materials (RMs) were prepared and distributed for PT schemes. The consensus values from the PT were calculated by robust statistics with its standard uncertainties. The relative bias between the reference values and the consensus values was used as one of the indicators for the determination of the assigned values. The relationship between the performances of the laboratories, expressed by *Z* scores, and the concentration levels of the RMs were also revealed.

Keywords Proficiency testing · Consensus values · *Z* score · Real water-based reference materials

Introduction

The analysis of environmental samples is quite challenging to ensure the traceability of the measurements since the environmental matrix may influence the determination, which cannot be estimated by matrix-free calibrants [1, 2]. Reference materials (RMs) with real environmental matrix played an important role in validating accuracies of analytical data. The participation of proficiency testing (PT) by using matrix-matched RMs has been recognized as an alternative approach for this purpose. Especially for regulatory purposes, it is often required to check the quality of environmental chemical analyses by routine PT. However, matrix-matched RMs are not readily available due to the cost of preparation of RMs [3].

In Republic of Korea, there are more than 250 laboratories performing environmental chemical analysis. The regulatory PT program organized by National Institute of Environmental Research, Ministry of Environment since 1984 consisted of five kinds of programs in wastewater, drinking water, solid waste, soil and indoor air analysis areas. Each PT program was performed once a year. In this study, two different kinds of real water-based RMs have been prepared by spiking of four trace metals such as lead, cadmium, chromium and copper. The reference values and uncertainty budgets of the RMs were determined via single laboratory and interlaboratory comparison approaches. The RMs were distributed for PT samples and the performance results of PT participants, expressed with *Z* scores, were evaluated using consensus values and targeted standard deviations.

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Y.-H. Kim (✉) · J.-S. Lee · K.-B. Song · Y.-Y. Kang · S.-K. Shin · G.-T. Jung · E.-J. Hong · H.-J. Kim · K.-H. Kim
Chemicals Behaviors Research Division, National Institute of Environmental Research, Ministry of Environment, Environmental Research Complex, Kyungseo-dong, Seo-gu, Incheon 404-708, Republic of Korea
e-mail: heek89@korea.kr

Results and discussion

Preparation of real water-based RMs

Two kinds of real water-based RMs were prepared using river water and lake water. The materials were sampled at River Han and Dae-Chung Lake, Republic of Korea in May and September 2008, respectively. For the preparation of

RMs, a 200 L-sized high-density polyethylene (HDPE) container was manufactured with a lid and a drain tap. It was also equipped with a magnetic stirrer (MSD50, Mi-Sung Co., Republic of Korea) for homogenization. About 300 L of water materials were collected and pre-acidified to ca. pH 2 with HNO_3 (Wako Chemicals, Japan). The samples were stored for 30 days for decanting suspended solids and filtered through 0.45 μm pore-sized membrane filter (Pall Corporation, AquaPrep 600 Capsules, PN. 12175, USA) and 0.2 μm pore-sized membrane filter (Whatman, POLYCAP 75 AS filter capsule, USA) by use of a peristaltic pump system to the pre-cleaned container. Targeted concentrations of trace metals such as lead, cadmium, chromium and copper were spiked with KRISS CRM standard solutions (PN. 105-02-PB2, 105-02-CD2, 105-02-CR2, 105-02-CU2, KRISS, Republic of Korea), followed by acidification with HNO_3 to pH 2. Homogenization of the materials was carried out in the container for 48 h by stirring with the magnetic stirrer and by recirculation with the pump which was connected to the drain tap with polyethylene piping. The homogenized materials were bottled in 250 mL portions to pre-cleaned HPDE bottles (PN. 382099-0250, Nalgene, USA) and each labeled bottles were weighed to check the change of the weight by transpiration. The materials were sterilized at 25 kGy by ^{60}Co radiation [4] and then sealed in polyethylene-coated aluminum zip-bags. The produced candidate RMs were stored in cold room at 4 °C. The RMs were designated as NIER-I08RW and NIER-I09LW.

Homogeneity and stability testing of RMs

The homogeneities of the candidate RMs were assessed by analysis of two test portions (about 50 g) of ten bottles selected randomly using inductively coupled plasma atomic emission spectrometry. Tables 1 and 2 show the reference values and uncertainty budgets for real water-based RMs, respectively. The uncertainties related homogeneities of within-bottle and between-bottle were evaluated by one-way ANOVA statistics as indicated by ISO Guide 35 [5]. The variances due to inhomogeneities were included in overall uncertainties of the reference values. Short-term and long-term stability testing was carried out at 40 and 25 °C for up to 6 months and the uncertainty constituent for stability was negligible and consequently was not included in the overall uncertainty [6]. Based on the findings obtained in the homogeneity study and the batch characterization, estimates for u_{bb} (homogeneity) and u_{char} (characterization) were obtained and combined to derive expended uncertainties with coverage factor of 2 according to the following equation:

$$U_{CRM} = 2\sqrt{u_{bb}^2 + u_{char}^2}. \quad (1)$$

Table 1 Uncertainty budget for river water-based RM, NIER-I08RW

Analytes	Reference value (mg/kg)	u_{bb}		u_{char}		U_{CRM}	
		mg/kg	%	mg/kg	%	mg/kg	%
Pb	0.628	0.004	0.6	0.006	0.9	0.014	2.2
Cd	0.061	0.0004	0.7	0.001	1.8	0.002	3.8
Cr	0.299	0.006	2.1	0.003	1.0	0.014	4.8
Cu	0.357	0.001	0.4	0.004	1.0	0.008	2.2

Table 2 Uncertainty budget for lake water-based RM, NIER-I09LW

Analytes	Reference value (mg/kg)	u_{bb}		u_{char}		U_{CRM}	
		mg/kg	%	mg/kg	%	mg/kg	%
Pb	1.236	0.004	0.4	0.009	0.8	0.020	1.8
Cd	0.486	0.003	0.6	0.003	0.7	0.008	2.0
Cr	1.016	0.004	0.4	0.006	0.6	0.014	1.4
Cu	0.828	0.006	0.8	0.006	0.7	0.016	2.0

Evaluation of consensus values from PT

Two kinds of the consensus values were assigned as the robust means of determined metal mass fractions obtained by governmental laboratories for expert laboratories ($n = 25$) and the robust means of determined metal mass fractions from PT schemes (Tables 3, 4) to evaluate the quality of the performances of PT. A total of 69 and 126 laboratories were attended for PT using NIER-I08RW and NIER-I09LW, respectively. All the data from the PT were normally distributed as shown in Fig. 1. The robust means of each consensus values were determined by "algorithm A" in ISO 13528 [7, 8] and the combined standard uncertainties of consensus values, u_c, were estimated according to following equation:

$$u_c = 1.25s^*/\sqrt{n} \quad (2)$$

where s^* is the robust standard deviation of the results calculated using algorithm A and n is the number of participating laboratories.

The consensus values from governmental laboratories and PT schemes were comparable and showed good correlations within the expended uncertainties. Thus, the robust means of PT results were selected as the consensus values in both PT schemes. It is noteworthy that the averaged standard deviations of NIER-I08RW were 6.9% by governmental laboratories and 9.8% by PT scheme, whereas there were no significant differences in the averaged standard deviations of NIER-I09LW between governmental laboratories (4.8%) and PT scheme (4.6%).

Table 3 Comparison of consensus values of NIER-I08RW

Analyte	By governmental lab ($n = 25$)			By proficiency testing ($n = 69$)		
	Robust mean (mg/kg)	Robust SD (mg/kg)	u_x (%)	Robust mean (mg/kg)	Robust SD (mg/kg)	u_x (%)
Pb	0.627	0.036	1.4	0.644	0.052	1.2
Cd	0.055	0.004	2.0	0.057	0.007	1.9
Cr	0.291	0.016	1.4	0.299	0.031	1.5
Cu	0.337	0.029	2.1	0.340	0.028	1.2

SD standard deviation

Table 4 Comparison of consensus values of NIER-I09LW

Analyte	By governmental lab ($n = 25$)			By proficiency testing ($n = 126$)		
	Robust mean (mg/kg)	Robust SD (mg/kg)	u_x (%)	Robust mean (mg/kg)	Robust SD (mg/kg)	u_x (%)
Pb	1.265	0.055	1.2	1.265	0.063	0.6
Cd	0.493	0.028	1.6	0.488	0.023	0.5
Cr	1.026	0.046	1.3	0.993	0.049	0.6
Cu	0.833	0.038	1.3	0.835	0.032	0.4

SD standard deviation

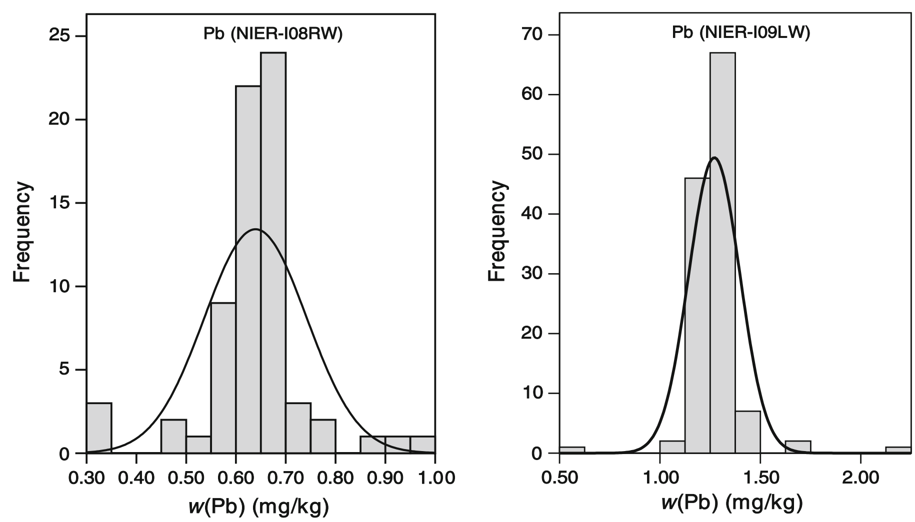

Fig. 1 Histogram of lead mass fraction, w(Pb), data from the PTs NIER-I08RW and NIER-I09LW

This is probably related with higher concentrations of targeted metals in NIER-I09LW than NIER-I08RW by 2–9 folds.

Determination of assigned values for PT

The participant's performances of the PT schemes were evaluated by Z scores in accordance with ISO 13528 (Eq. 3):

$$Z = \frac{X_i - X_s}{\sigma^*}. \tag{3}$$

The assigned values, X_s, were used as the reference values or the consensus values to compare the performance results of PT schemes based on the reference values and those based on the consensus values. The targeted standard deviation, σ^*, was set as $0.05X_s$ according to the limit of 15% on the deviation in Korean Environmental Water Quality Standard. The targeted standard deviations did not show any significant differences with robust standard deviations in PT scheme using NIER-I09LW. The relative bias, B_{relative}, was calculated to see the deviations between the reference values and the consensus values according to following equation [9]:

$$B_{\text{relative}} = \frac{X_{\text{cons}} - X_{\text{ref}}}{\sigma^*} \tag{4}$$

The relative bias, B_{relative}, enabled a comparison of the reference values and the consensus values. In Tables 5 and 6, B_{relative} and the number of the laboratories categorized by Z scores were given to evaluate the PT performances by different assigned values. Out of a total of 1552 results,

Table 5 Comparison of the number of laboratories categorized by Z scores in the PT study (NIER-I08RW)

Items	Pb		Cd		Cr		Cu	
	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.
$\lvert Z \rvert \geq 3$	13	11	14	15	17	17	14	11
$2 \leq \lvert Z \rvert < 3$	3	6	12	11	15	11	7	10
$\lvert Z \rvert \leq 2$	53	52	43	43	37	41	48	48
B_{relative}	0.50		0.05		0.12		−0.93	

Table 6 Comparison of the number of laboratories categorized by Z scores in the PT study (NIER-I09LW)

Items	Pb		Cd		Cr		Cu	
	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.	Ref.	Cons.
$\lvert Z \rvert \geq 3$	6	5	15	13	6	5	9	8
$2 \leq \lvert Z \rvert < 3$	9	7	9	7	16	15	7	10
$\lvert Z \rvert \leq 2$	111	114	102	106	104	106	110	108
B_{relative}	0.46		0.08		−0.45		0.18	

only 46 of them (2.9%) were depended on the assigned values used. Thus, in this study, the reference values were used as assigned values since the bias, B_{relative}, was less than ± 1 and both values were in a good agreement.

The average results of the PT performances for NIER-I08RW, obtained via the reference values, showed that out of 69 laboratories, 13 were unsatisfactory (21%) and 10 were questionable (13%). However, the average results of the PT performances for NIER-I09LW showed that out of 126 laboratories attended, only 8 were unsatisfactory (7%) and 10 were questionable (8%) and implied that the majority of Korean environmental laboratories showed relatively good performances in water analysis. It is interesting to compare the performance results in two different PT schemes. The portion of the laboratories evaluating as "in agreement" was dramatically dropped from 85 to 67% by decreasing the concentration levels of NIER-I09LW to those of NIER-I08RW, whereas the portion of the laboratories evaluating as "questionable" and "unsatisfactory" was increased by more than twofold from 15 to 34%. These differences between two PT schemes implied that the concentration level of the RMs impacted the performances of the laboratories.

Conclusion

In order to evaluate the performances of Korean environmental laboratories operating in the field of water analysis, real water-based PT schemes have been organized by National Institute of Environmental Research, Ministry of Environment. In this study, two different kinds of RMs, NIER-I08RW and NIER-I09LW, for trace metal analysis were prepared and distributed for the PT schemes to 69 and 126 laboratories, respectively. The consensus values were obtained from the robust means of the PT results after minimizing the influence of outliers by the use of the robust statistical procedure by ISO 13528. The assigned values for the PT schemes were determined as the reference values since the consensus values and the reference values were relatively less biased. The targeted standard deviation, σ^*, was set as $0.05X_{\mathrm{s}}$ according to the limit of 15% on the deviation in Korean Environmental Water Quality Standard. From the comparison of the Z scores of two different PT schemes, it is revealed that concentration level of the RMs impacted the performances of the laboratories. Thus, the concentration levels of RMs should be carefully chosen to accomplish the purpose of PT scheme. Currently, more PT studies using real matrix-based RMs such as soil and fly ashes are ongoing to improve the competences of Korean environmental laboratories in trace metal analysis.

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